

Trade and Financial Integration: A Blessing or a Curse?

A Network Perspective

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Research Question

Is trade and financial integration a blessing or a curse for
economic growth?

- Blessing
 - Expanded market access
 - Cheap credit (exorbitant privilege)
 - UK (1760s-1840s): “Workshop of the world”
- “Financial Resource Curse” (Benigno and Fornaro, 2014; Benigno, Fornaro, and Wolf, 2025)
 - Diversion from manufacturing to nontradable sector (exorbitant burden)
 - Reduced incentives to innovate
 - US

Key Conditions for a “Curse”

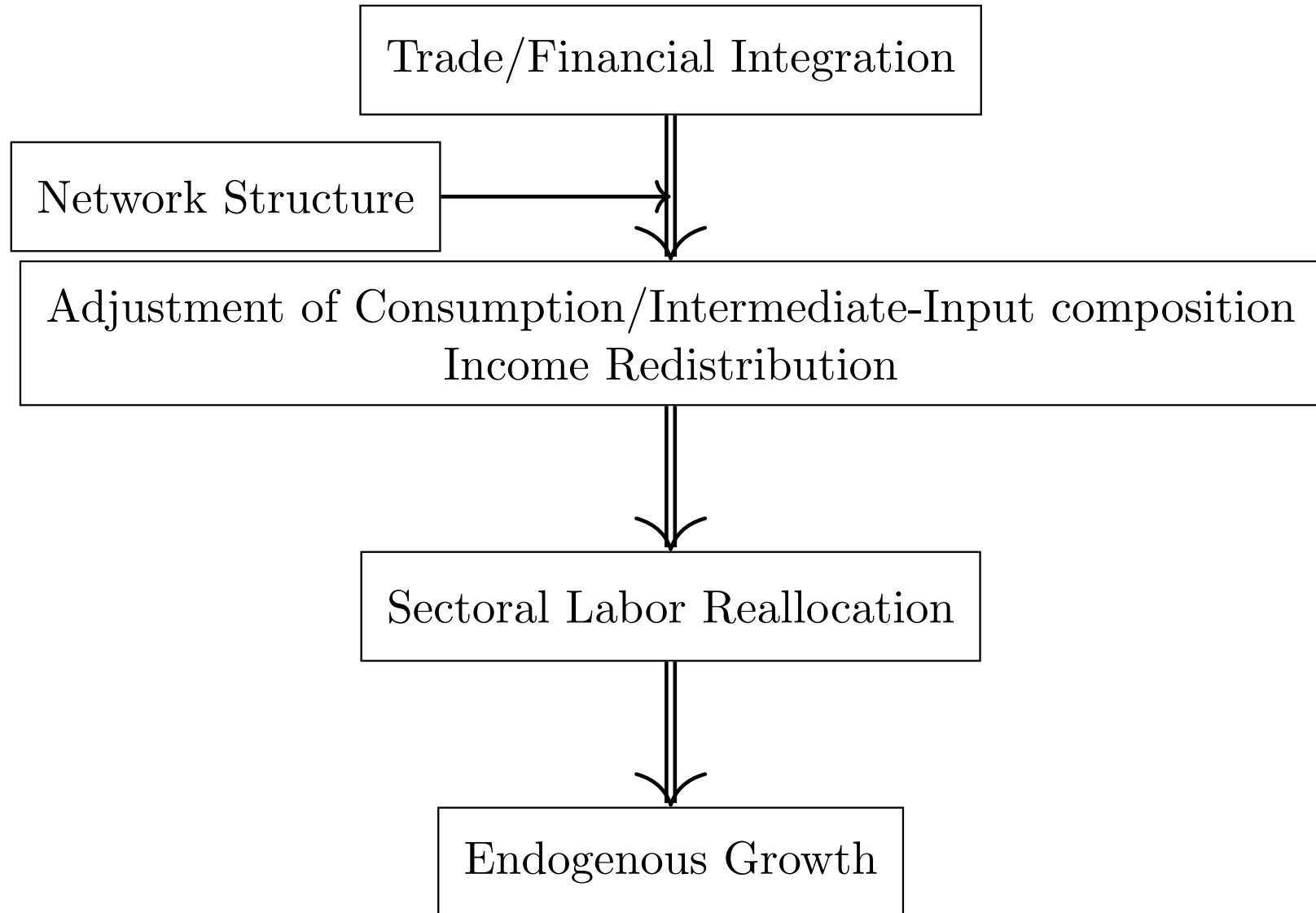
Beningo et al. (2025)

- **No production network**
 - Absent downstream sectors for the “engine” sector
- **Limited to two sectors**
 - Absent rich sectoral re-allocation patterns
- **Financial distortion by trade partner**
 - Sustained trade deficit

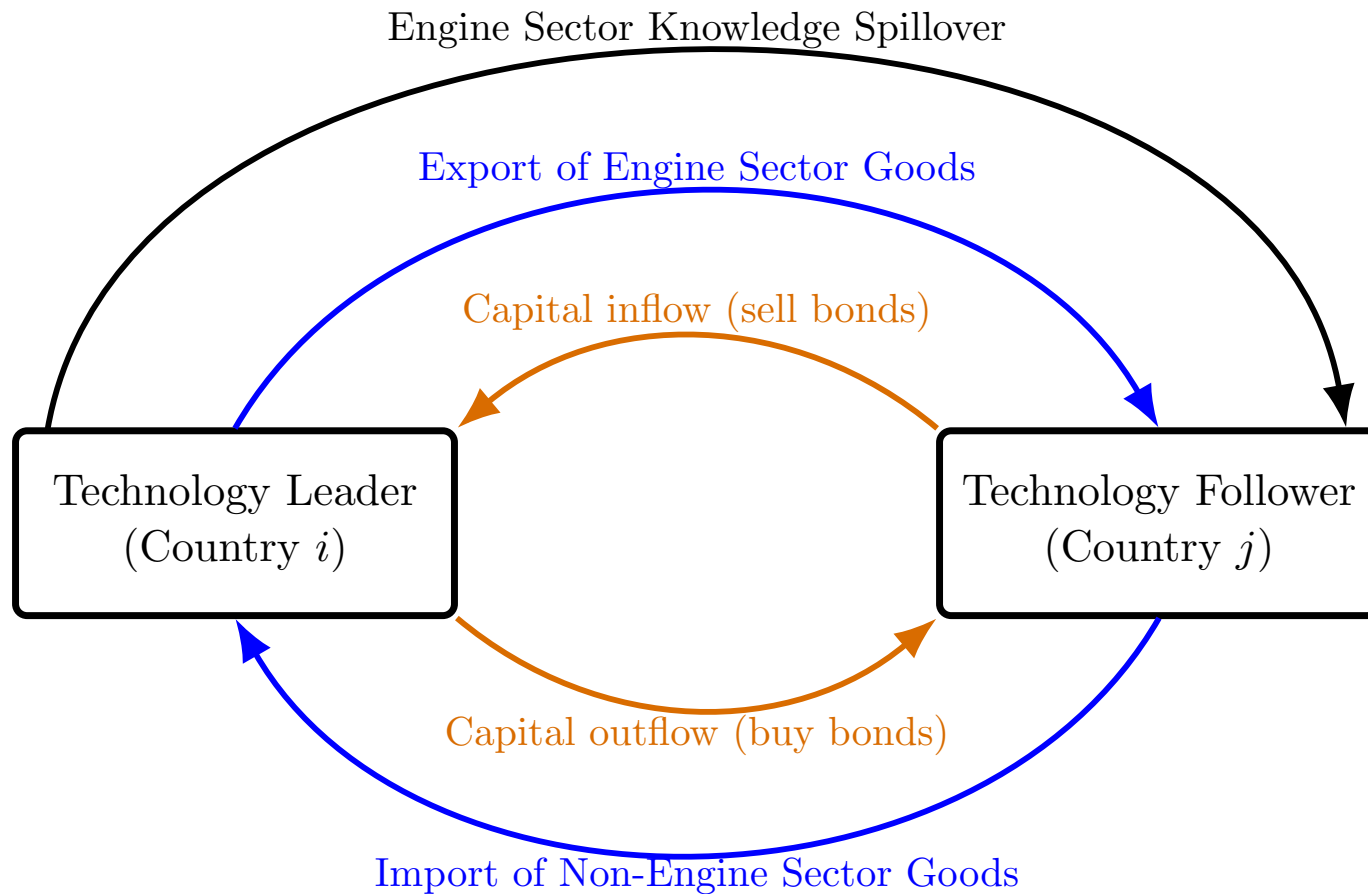
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Key Mechanism



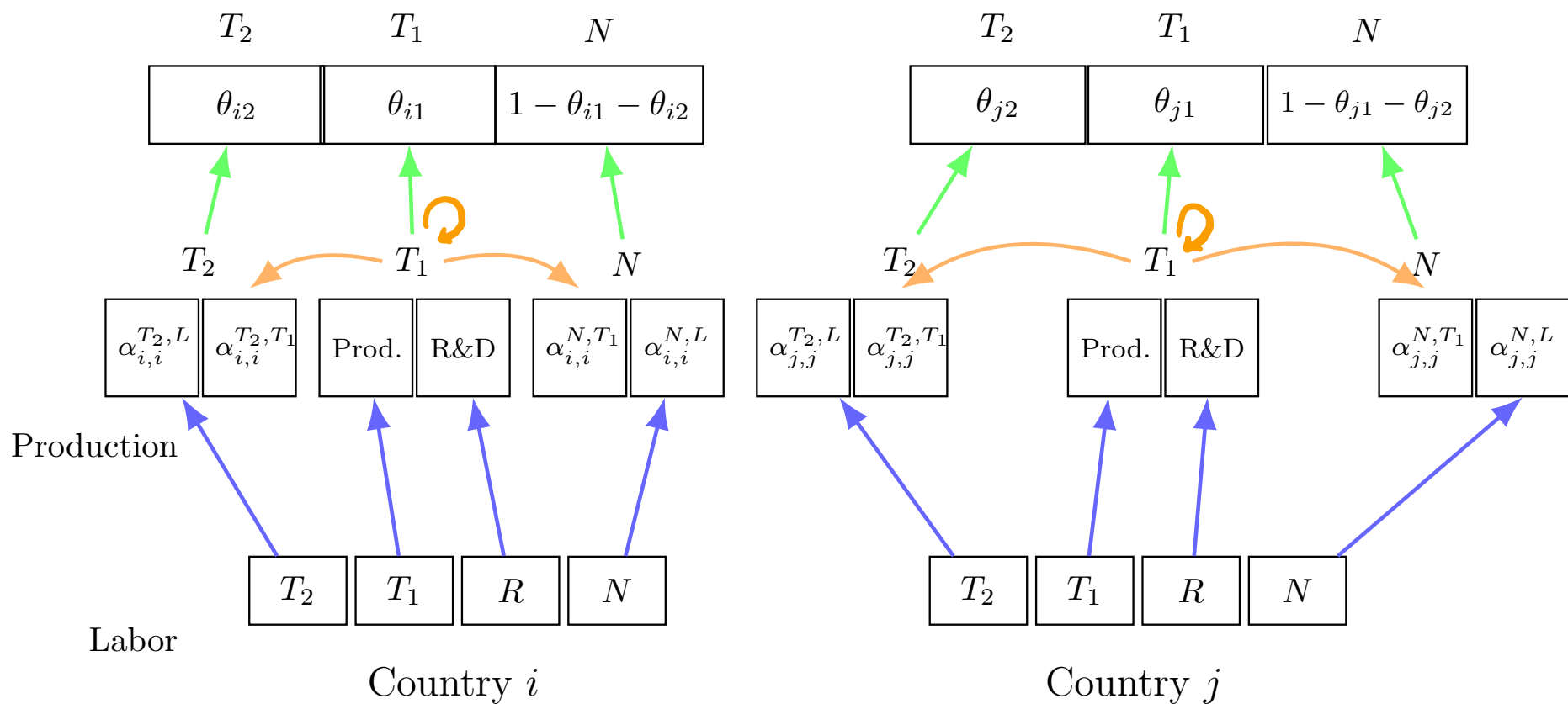
Flow Diagram of the Model



- Imports – Exports = Financial inflow
- L_i^{T1} determines global economic growth
- L_j^{T1} determines gap from frontier

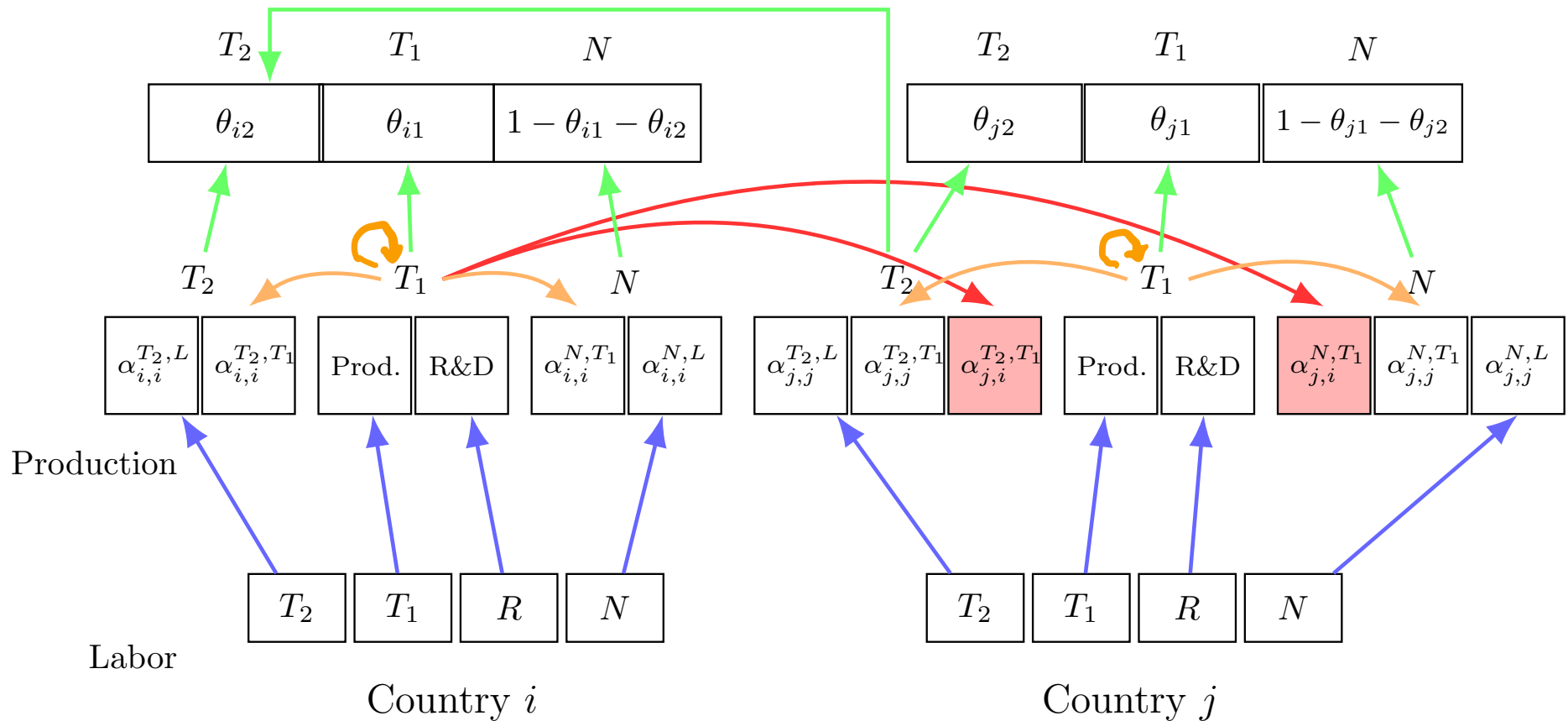
Cobb-Douglas Utility and Production Functions

Consumption



Cobb-Douglas Utility and Production Functions

Consumption



Blessing or Curse

when Country j uses Financial Distortion

- Country i expands trade deficit at $zA_{i,t}$, where $z \propto$ borrowing limit.
- Trade Deficit

$$P_t^{T2} M_{i,j,t} - Exports = zA_{i,t}$$

- What is new?
 - Country i 's imports not fully supported by exports
 - Income effect:
 - negative for country j (financial outflow)
 - positive for country i (financial inflow of $zA_{i,t}$)

Financial Distortion: Model III

- Import value

$$P_t^{T2} M_{i,j,t} = H P_{j,t}^{T1} C_{j,t}^{T1} + \frac{z A_{i,t}}{1 - \alpha_{j,i}^{T2,T1}}$$

where H proxies consumption-weighted centrality of country i 's T1 goods in country j

$$H = \frac{\alpha_{j,i}^{T2,T1} \frac{\theta_{j,2}}{\theta_{j,1}} + \alpha_{j,i}^{N,T1} \frac{1 - \theta_{j,1} - \theta_{j,2}}{\theta_{j,1}}}{1 - \alpha_{j,i}^{T2,T1}}$$

- Importance of $\alpha_{j,i}^{T2,T1}$:
 - Global direct and indirect impact on production and consumption
 - Multiplier effect: Country j has to import more to export more (counteracting distortion)
 - A measure of domestic value-added captured in foreign exports

Ambiguous Change in Demand in Country i

- Net export demand for T1 is given by

$$\begin{aligned}
 & \underbrace{(P_t^{T2} M_{i,j,t} - z A_{i,t})}_{\text{Exports}} - \underbrace{\alpha_{i,i}^{T2,T1} P_t^{T2} M_{i,j,t}}_{\text{Otherwise domestically produced}} \\
 = & \alpha^{T2,L} H P_{j,t}^{T1} C_{j,t}^{T1} + \left(\alpha_{j,i}^{T2,T1} - \underbrace{\alpha_{i,i}^{T2,T1}}_{\text{non-labor share}} \right) \frac{z A_{i,t}}{1 - \alpha_{j,i}^{T2,T1}}
 \end{aligned}$$

- Net demand for T1 depends upon
 - Labor share in T2 sector
 - $\alpha_{j,i}^{T2,T1}$ close to the non-labor share
 - Country j 's exports embodied primarily the value of domestic T1 goods

Labor Market Equilibrium in Country i - Model III

Labor market equilibrium condition along balanced growth:

$$\bar{L} + k_i \frac{(L_j^{T1})^{\gamma+1}}{(g - \beta)^\gamma} + z f(\Phi) = (1 + F_i^{T2, aut} + F_i^{N, aut}) L_i^{T1} + \frac{g - 1}{\chi}$$

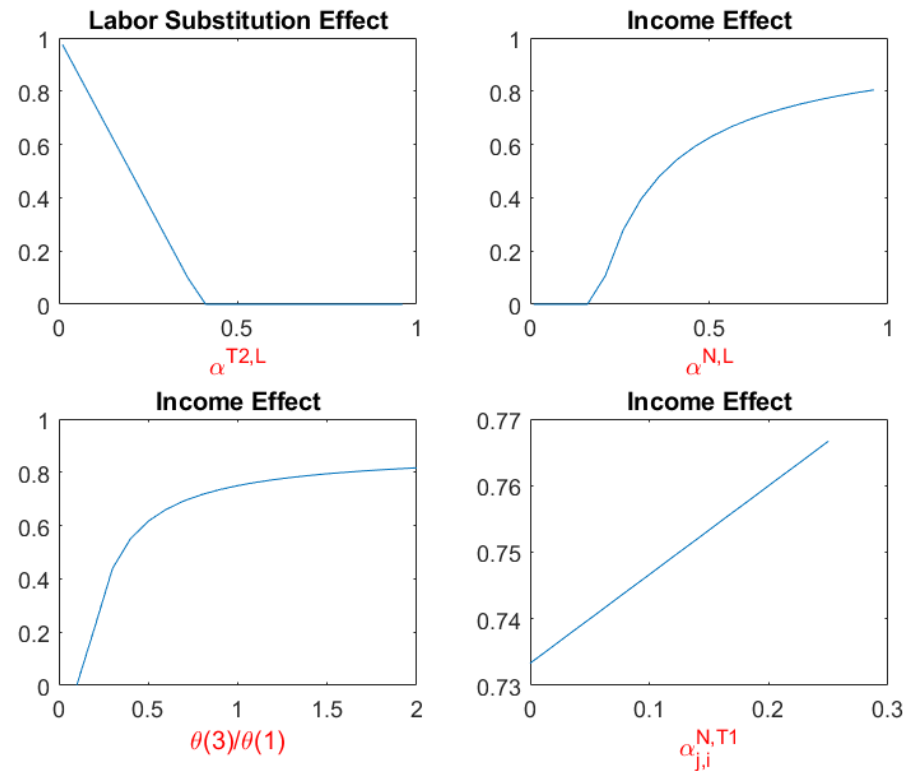
- $f(\Phi)$ more likely to be positive
 - the more central Country i 's T1 in Country j 's T2 sector (limiting import substitution), that is the larger $\alpha_{j,i}^{T2,T1}$ is relative to $\alpha_{i,i}^{T2,T1}$ (or $\alpha_{j,j}^{T2,T1}$)
 - the larger the labor share in T2 sector (more labor release)
 - the more favorable the income effect
 - The smaller the consumption-weighted compensation to labor in N relative to T1.

Model Parameters

Variable	Value	Interpretation
$\alpha^{T2,L}$	$\frac{2}{3}$	Labor share in T2
$\alpha_{j,i}^{T2,T1}$	$\frac{1}{6}$	exponent in sector T2 country j
$\Theta(k, s)$	$\frac{1}{3}$	final demand composition
$\bar{L}_i, \bar{L}_j$	1	Labor endowment
$\alpha_{i,i}^{N,L}, \alpha_{j,j}^{N,L}$	$\frac{3}{4}$	Labor shares in sector N
$\alpha_{j,i}^{N,T1}$	$\frac{1}{2}(1 - \frac{3}{4})$	exponent in sector N country j
β	0.96	Subjective discount rate
α	0.122	Exponent for intermediate (innovation) sector
χ	3.02	Productivity of research in country i
ξ	2	Productivity of research in country j
ϕ	0.5	Contribution of country i 's knowledge
z	0.05	proportional borrowing limit

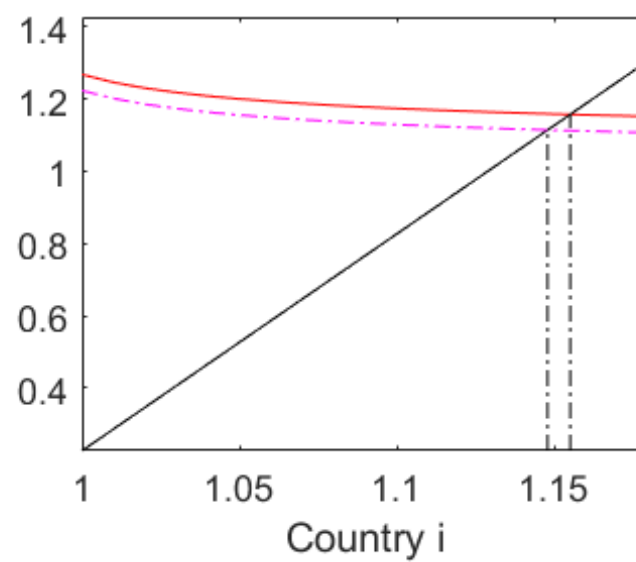
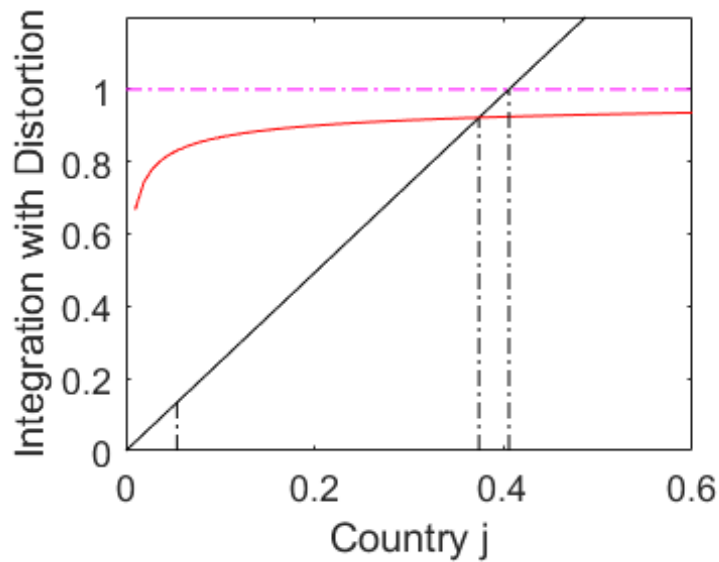
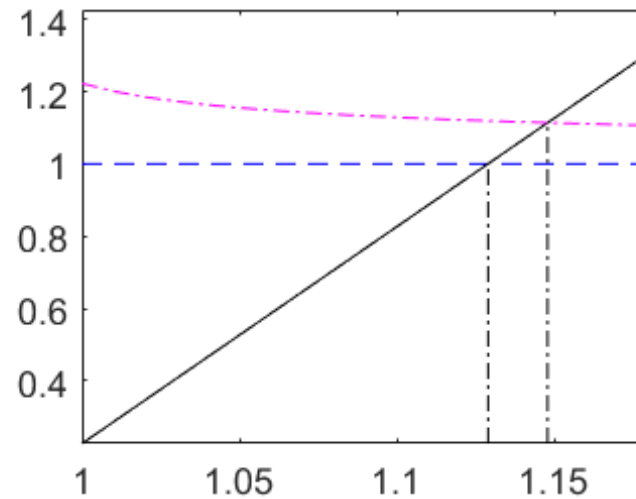
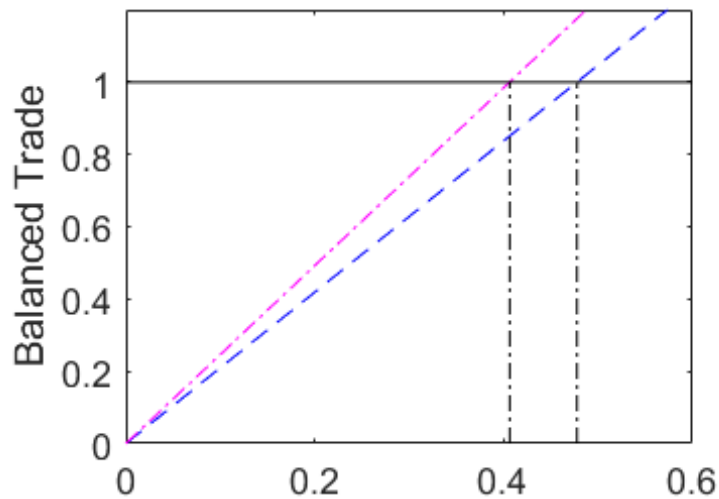
Threshold Defined by $\alpha_{j,i}^{T2,T1}$

Fundamentally, “Importing” labor or Country j ’s T1 goods through importing T2?

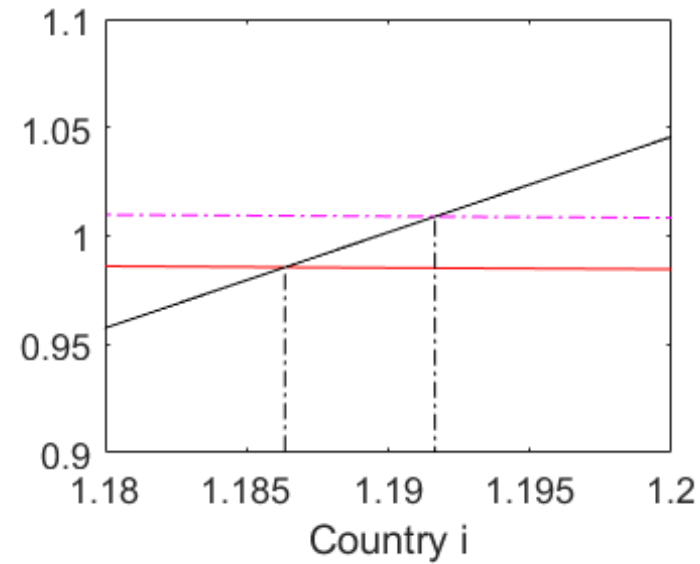
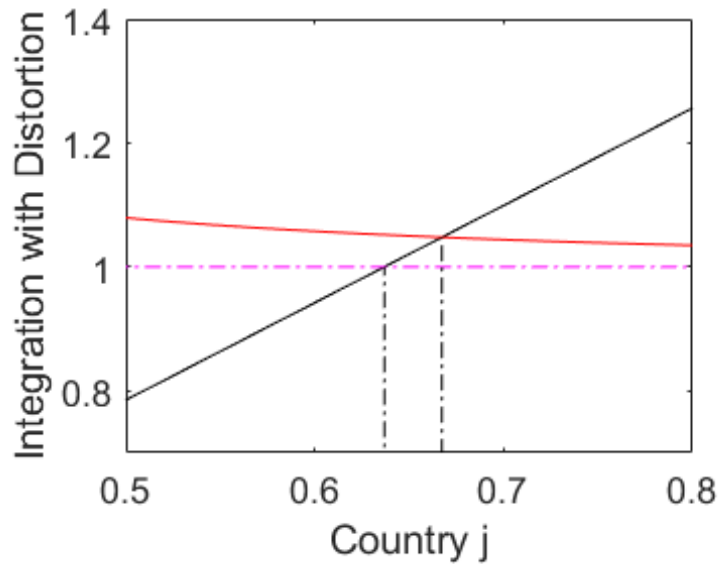
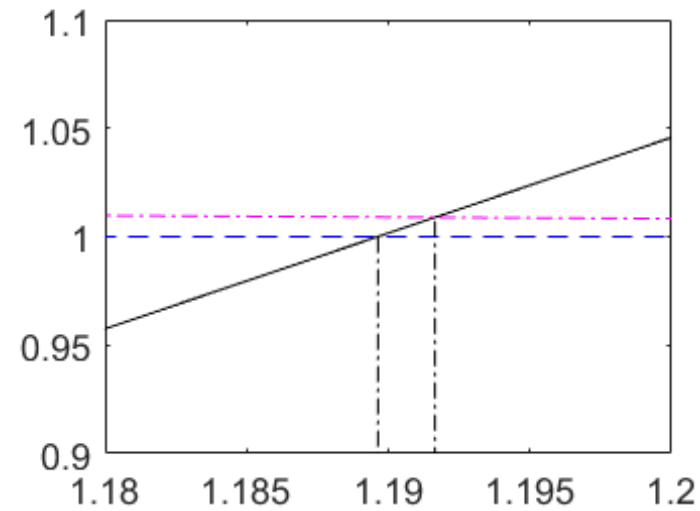
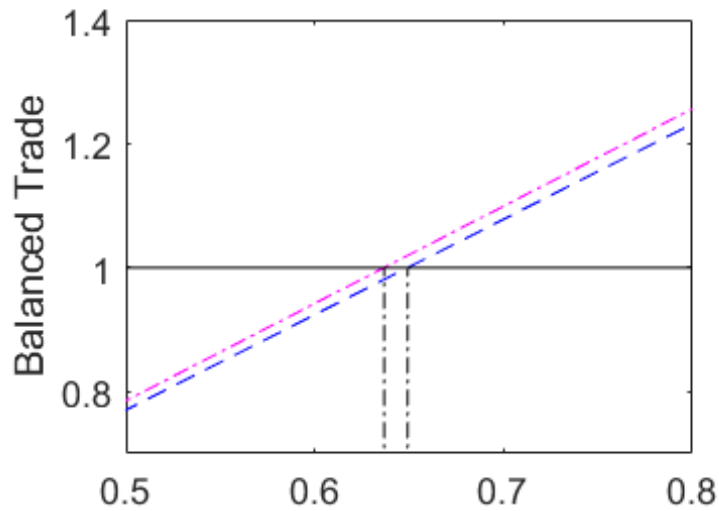


For plots 2-4, $\alpha^{T2,L} = 0.1$, all other parameters at benchmark value

Benchmark Case: $\alpha^{T2,L} = \frac{2}{3}$, $\alpha_{j,i}^{T2,T1} = \frac{1}{6}$



A Case of Curse: $\alpha^{T2,L} = 0.1, \alpha_{j,i}^{T2,T1} = 0.2$



Numerical Results

	Blessing			Curse		
	Autarky	Integ	Dist	Autarky	Integ	Dist
g	1.1290	1.1479	1.1551	1.1896	1.1916	1.1864

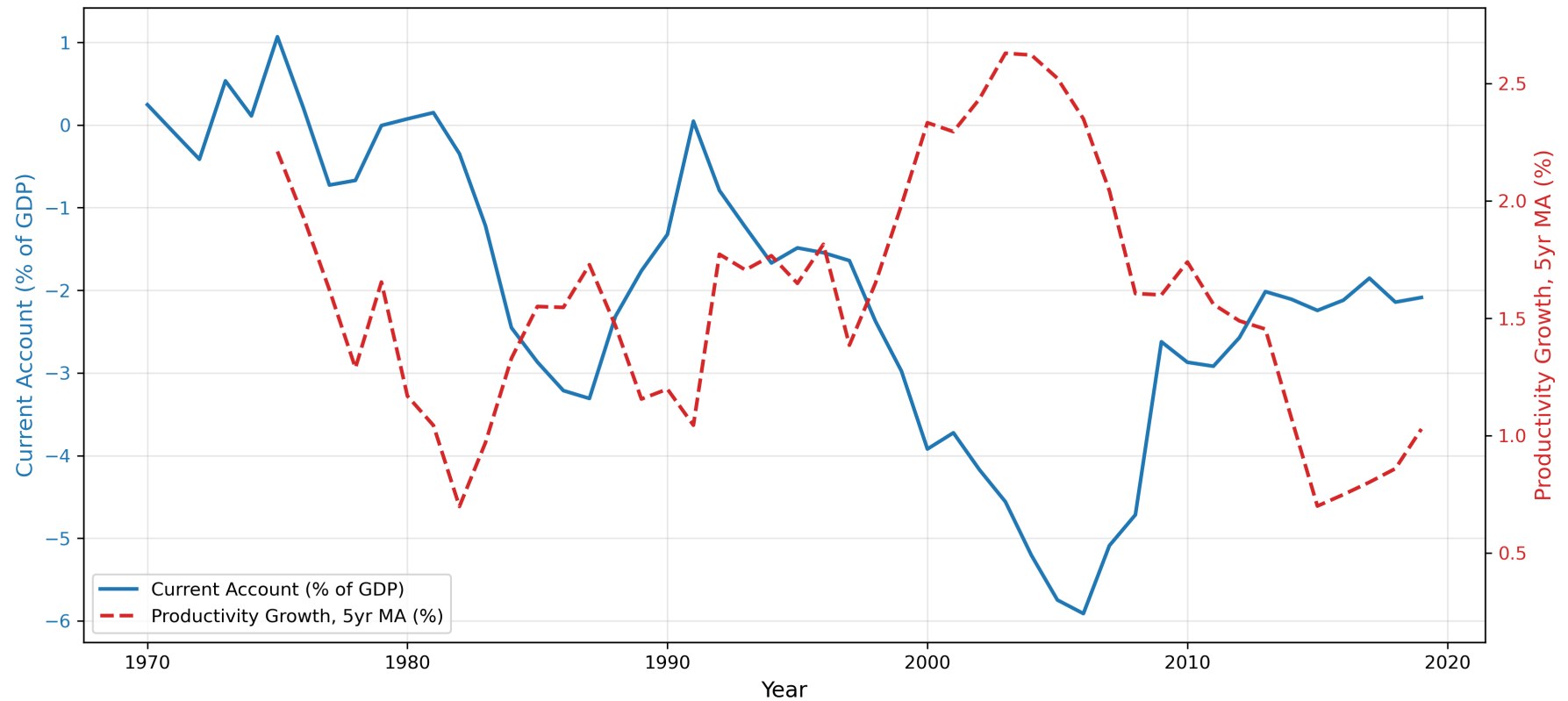
Table: Autarky, Integration, and Distortion for Blessing vs Curse

Coun	Autarky				Integration				Distortion			
	T1	T2	N	R	T1	T2	N	R	T1	T2	N	R
i	0.478	0.226	0.254	0.043	0.531	0.166	0.254	0.049	0.552	0.118	0.280	0.051
j	0.478	0.226	0.254	0.043	0.406	0.304	0.253	0.037	0.374	0.383	0.208	0.035
i	0.649	0.034	0.254	0.063	0.655	0.028	0.254	0.063	0.640	0.019	0.280	0.062
j	0.649	0.034	0.254	0.063	0.637	0.048	0.254	0.062	0.667	0.155	0.114	0.064

Table: Labor Allocations - Blessing (Top) vs Curse (Bottom)

Time Series Pattern

US Current Account and Productivity Growth (1970-2019)



Policy Implications

- Both countries can implement trade and industrial policies
 - Subsidize production costs of engine sector
 - Subsidize exports
- Policies have growth implications through resource allocations.
- Leap-frog possible when country j is or becomes more research productive.

Conclusion

- We construct a model combining global production network, innovation-driven endogenous growth and trade-financial integration
- Trade and financial integration can be either blessing or curse, depending upon
 - Centrality of the Engine sector of the technology leader (determining the competition between two engine sectors)
 - Labor intensity and demand structure
 - Characteristics of trade-financial integration
- Implications for trade, financial and industrial policies.